EIN3241 – Human Factors & Ergonomics

Quiz 3 - Dr. Jad Atweh

Part A: Multiple Choice Questions (6 points)

1. During the onset of physical work, heart rate initially lags oxygen demand, creating:
   a) Heat stroke
   b) Cardiovascular plateau
   c) Lactate threshold
   d) Oxygen recovery
   e) Oxygen debt

2. Which contraction type restricts blood flow and accelerates fatigue?
   a) Dynamic
   b) Static
   c) Isokinetic
   d) Alternating

3. If two individuals perform the same task and one shows a lower heart rate response, that person is likely:
   a) Overexerting
   b) Less fit
   c) More aerobically conditioned
   d) Experiencing oxygen debt
   e) Working at a higher MAP

4. Kinematics focuses on:
   a) Forces and torques
   b) Muscle activation
   c) Description of motion only
   d) Internal joint stress
   e) Joint compression forces

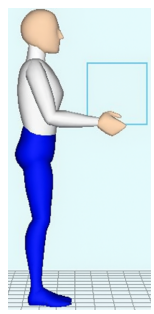
5. A sagittal plane divides the body into:
   a) Upper and lower halves
   b) Front and back halves
   c) Left and right halves
   d) Medial and lateral halves
   e) Proximal and distal halves

6. Abduction refers to movement:
   a) Toward the midline
   b) Away from the midline
   c) Around a long axis
   d) Increasing joint torque
   e) Decreasing moment arm

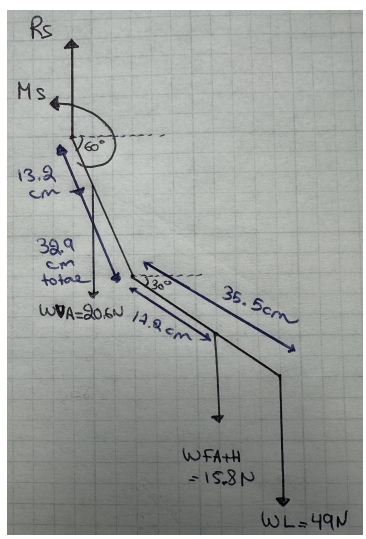
Part B: Exercise (14 points)

A person holds a load with one hand with the arm bent. The upper arm makes an angle of 60° with the horizontal and the forearm makes an angle of 30° with the horizontal. All weights act vertically downward at their centers of mass.

- $W_{UA} = 20.6$ N acts at the upper arm center of mass, located 13.2 cm from the shoulder along the upper arm.
- $W_{FA+H} = 15.8$ N acts at the forearm+hand center of mass, located 17.2 cm from the elbow along the forearm.
- $W_{load} = 49$ N acts at the hand center of mass, located 35.5 cm from the elbow along the forearm.
- The elbow is 32.9 cm from the shoulder along the upper arm.



Draw the free-body diagram and calculate the Shoulder reaction force and the Shoulder reaction moment using the **one-segment approach**. If you use the segment-to-segment approach, you will receive **no credit**.



$$\sum F_y = 0: \quad F_{\text{shoulder}} - W_{\text{UA}} - W_{\text{FA+H}} - W_{\text{load}} = 0$$

$$F_{\text{shoulder}} = 20.6 + 15.8 + 49 = 85.4\,\text{N (upward)}.$$

$$\sum M_{\text{shoulder}} = 0: \quad M_{\text{shoulder}} - \Big(W_{\text{UA}} d_{\text{UA}} + W_{\text{FA+H}} d_{\text{FA}} + W_{\text{load}} d_{\text{load}}\Big) = 0$$

$$M_{\text{shoulder}} = 49\big(35.5\cos 30^\circ + 32.9\cos 60^\circ\big) + 15.8\big(17.2\cos 30^\circ + 32.9\cos 60^\circ\big) + 20.6\big(13.2\cos 60^\circ\big)$$

$$M_{\text{shoulder}} = 2312.5 + 495.26 + 135.96 = 2943.72\,\text{Ncm} \approx 29.44\,\text{Nm (CCW)}.$$